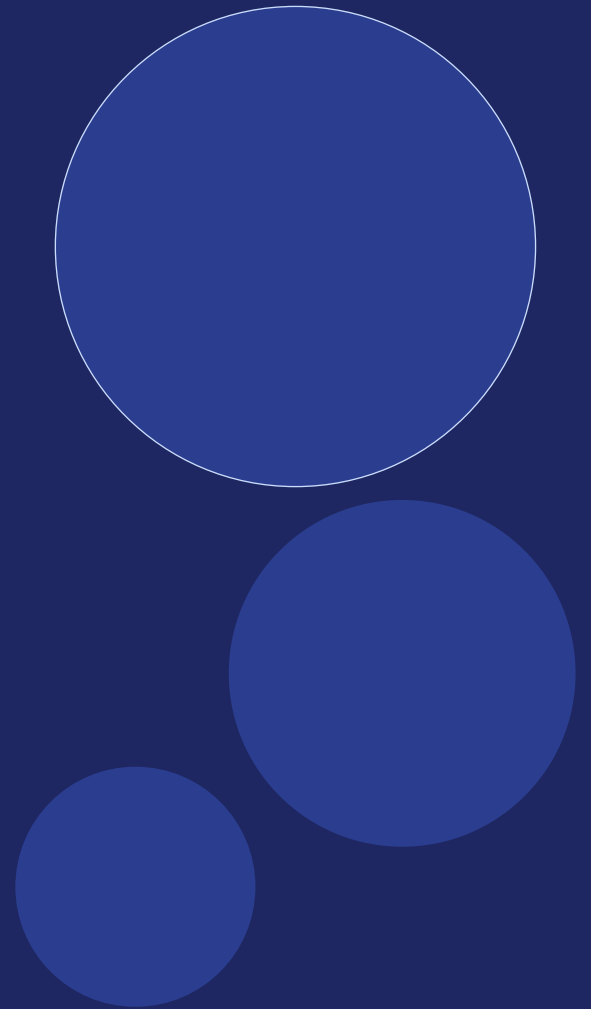


Intelligent Candidate Ranking

Beyond Keywords: How a Great Recruiter Really Reads a Profile

InnoCoders | Redrob Hackathon 2025



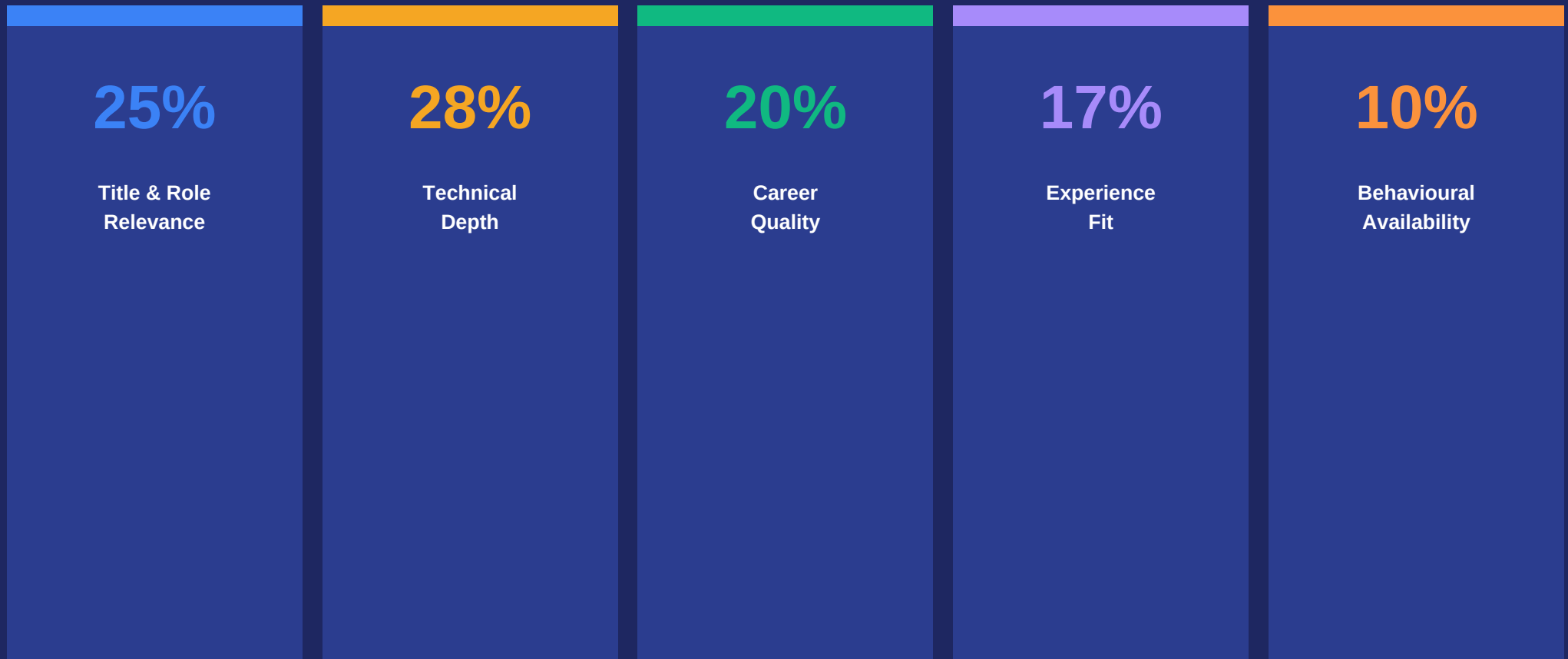
Why Keyword Filters Fail

THE TRAP: A Marketing Manager with PyTorch, FAISS, LLM in skills ranks ahead of an ML Engineer with 7 years of production retrieval work — because they have more matching keywords.

- ✗ Keyword density $\neq$ domain expertise — the JD explicitly calls this out as a trap
- ✗ Skills section is self-reported and trivial to game — anyone can list FAISS
- ✗ A perfect-on-paper candidate inactive for 6 months is, for hiring, not available
- ✗ Great recruiters read career arcs, not skill checklists

Our Approach: Five-Dimensional Scoring

Reading the profile like a great recruiter would



Location is a soft multiplier ($\times 0.70$ – $\times 1.00$, Pune/Noida preferred). Hard disqualifier penalties stack multiplicatively on top.

Technical Depth: Catching the Keyword Stuffer

**KEY INSIGHT: Skills claimed must be backed by career evidence.
Career descriptions are the ground truth — the hardest signal to fake.**

- 45%** Career descriptions weighted at 45% of tech score — hardest signal to fake
- 35%** Skill list contributes 35%, but cross-validated against career descriptions
- 20%** Platform assessment scores provide independent 3rd-party validation

CONSISTENCY PENALTY RULES

- ≥ 3 advanced ML skills + zero description evidence → $\text{skill} \times 0.25$
- ≥ 3 advanced ML skills + weak evidence → $\text{skill} \times 0.55$
- Skills with endorsements unsupported by role history → discounted

EXAMPLE: FAISS listed as 'advanced' by a Frontend Engineer whose career is entirely React / QA / Android.
No FAISS in any job description. Skill score discounted by 75%.

Title Tier Taxonomy

Current title is the strongest single signal — career-peak also counts (65% / 35% split)

TIER 1

1.00

TIER 1 — Direct examples:

ML Engineer, NLP Engineer, Search Engineer, Recommendation Engineer, Applied ML Engineer, Applied Scientist

TIER 2

0.75

TIER 2 — Direct examples:

Data Scientist, Senior Data Scientist, AI Research Engineer, AI Specialist, MLOps Engineer

TIER 3

0.50

TIER 3 — Direct examples:

Software Engineer, Data Engineer, Backend Engineer, Senior Software Engineer, Full Stack Developer

DISQUAL

×0.25

DISQUAL — Direct examples:

Marketing Manager, HR Manager, Accountant, Civil Engineer, Mechanical Engineer, Sales Executive

Explicit Disqualifiers from the JD

×0.25

Irrelevant current title

HR / Marketing / Accountant / Sales Executive etc.

×0.50

Entire career in consulting

>85% tenure at TCS / Infosys / Wipro / Accenture / Cognizant / HCL

×0.60

Ghost candidate

Inactive 6+ months AND open_to_work_flag is false

×0.55

CV/Speech without NLP

Computer Vision or Speech Engineer with zero NLP/retrieval crossover

×0.85

Long notice period

Notice > 90 days (JD prefers ≤30 days, can buy out up to 30)

All penalties are multiplicative. A ghost Marketing Manager at a consulting firm could receive $\times 0.25 \times \times 0.50 \times \times 0.60 = \times 0.075$ final multiplier.

Honeypot Defense: Internal-Consistency Verification

The dataset plants "honeypot" profiles that look perfect but are subtly impossible.
We verify every profile against itself before it can be ranked.

Date-arithmetic check

Role claims 166 months of tenure but its own start date is 33 months ago - duration contradicts the dates.

Expert-with-zero-use check

Skills claimed at 'expert' proficiency with 0 months of actual use - three or more flags the profile.

Experience-span check

Stated years-of-experience exceeds the entire span of the career history by 2+ years.

Technology-anachronism check

Claims 94 months of RAG experience - but RAG has existed only since May 2020. Month-precision dates for 15 AI technologies.

Result: 141 internally inconsistent profiles detected across 100K candidates - zero appear in our top 100.

Behavioural Signals: The Hiring-Reachability Layer

The best resume means nothing if they don't reply or aren't available

25%

Open to Work Flag

Are they actively looking? Highest single weight.

20%

Activity Recency

Last login < 30d = full. > 6 months = near-zero.

20%

Recruiter Response Rate

Direct predictor of whether they will engage.

15%

Notice Period

JD prefers ≤ 30 days. > 90 days $\rightarrow$ $\times 0.85$ global penalty.

10%

Interview Completion Rate

Reliability signal — attend scheduled interviews?

10%

GitHub Activity Score

Engineering engagement — active open-source contributors.

Results: Top 10 Shortlisted Candidates

Rank	Title	YOE	Location	Score
1	Senior Machine Learning Engineer	6.1 yrs	Pune	1.000
2	Recommendation Systems Engineer	8.0 yrs	Noida	0.868
3	Recommendation Systems Engineer	6.5 yrs	Pune	0.750
4	Applied ML Engineer	7.1 yrs	Gurgaon	0.718
5	Search Engineer	7.6 yrs	Gurgaon	0.700
6	Applied ML Engineer	5.7 yrs	Mumbai	0.651
7	Recommendation Systems Engineer	4.1 yrs	Pune	0.648
8	Lead AI Engineer	6.7 yrs	Jaipur	0.598
9	Senior Data Scientist	7.4 yrs	Delhi	0.597
10	Search Engineer	4.2 yrs	Mumbai	0.596

✓ 100% of top 100 have ML / AI / Search / NLP titles. Zero honeypots (141 inconsistent profiles detected and excluded).

Deliberate Design Choices

What We Did NOT Do

- ✗ Did NOT use keyword count as primary signal — the JD warns this is the trap
- ✗ Did NOT rank by skill quantity — quantity without career evidence is noise
- ✗ Did NOT use an LLM API — must run offline, CPU-only, no network
- ✗ Did NOT reward non-technical titles with AI keywords — ×0.25 penalty

What We DID Do Instead

- ✓ Use career description text as primary signal — hardest to fake
- ✓ Cross-validate skills against actual job descriptions for consistency
- ✓ Apply JD's stated disqualifiers as hard multiplicative penalties
- ✓ Weight recency of ML titles — stale 'architects' penalised

Production-Ready Design & Reproducibility

31s

Runtime for 100K candidates
(single CPU core)

0

Pip dependencies and
honeypots in shortlist

141

Internally inconsistent
profiles auto-detected

- ✓ Fully deterministic — no randomness, no API calls, no GPU required
- ✓ Pure Python standard library — zero pip install steps
- ✓ Single command: `python rank.py --candidates ./candidates.jsonl --out ./submission.csv`
- ✓ Passes `validate_submission.py` with zero errors
- ✓ Weights documented and easy to recalibrate for a different JD